

Week 3

Limited Retention Budget

Build a decision model that respects money, leakage, and operational limits.

BUILD

BREAK

DEFEND

Why this week matters

The practical motivation

- A high score is not a decision.
- A limited budget forces the model to answer: who gets the next dollar?
- The best teaching moment is a model that looks brilliant because it cheated.

The business can afford 8,000 retention offers, not 80,000. A model is only useful if it chooses better than a simple rule.

The studio rhythm

1

BUILD

Use the agent to create the first useful version.

2

BREAK

Attack the draft with bad data, edge cases, and misuse.

3

DEFEND

Keep only what survives evidence, tests, and explanation.

Mission brief

Use the agent to build baselines, a candidate model, and a cost-aware decision memo.

You will build

Baseline comparison, leakage audit, threshold rationale, and one-page decision memo.

Success looks like

- The artefact runs from a clean checkout.
- At least one agent failure is found and fixed.
- The final claim has evidence a peer can inspect.

Start here

Use these files and assumptions before asking the agent to build.

Starter materials

- Starter_Kit/sample_data/week3_retention_training.csv
- Sample_Data_and_Scenarios.md cost assumptions
- Starter_Kit/templates/defence_note_template.md
- Portfolio_and_Assessment_Model.md rubric

Confirm first

- What decision will the model change?
- Which features are unavailable at decision time?
- What capacity or budget limit sets the threshold?

Worked example to imitate

Use `Worked_Examples.md` and `Starter_Kit/examples/` for the full artefact.

What strong evidence looks like

- Good leakage audit: excludes target and post-offer response.
- Good baseline table: compares model with simple business rules.
- Good decision memo: ties threshold to cost, capacity, and limits.

Discuss before building

- What makes this example inspectable?
- Which claim is supported by evidence?
- What limit is still named?

Concept burst

Only the ideas needed for today's build

1

Baselines before models.

2

Leakage audit before celebration.

3

Threshold as a business decision.

Guided practice menu

Use the matching file in Exercises/ for timing, instructions, and instructor notes.

Core exercises

- Decision frame: action, eligibility, capacity, cost, and value.
- Leakage hunt: label every column by decision-time availability.
- Threshold economics: choose a top-N or cutoff policy.

Submit by the end

- One concrete artefact.
- One reproduced or plausible failure.
- One evidence-linked defence note.

Three-hour delivery path

Use these slides as pacing support: set context, practise, build, break, repair, defend.

Week 3

180-minute run of show

- 1 0-5: retrieval link and setup check.
- 2 5-15: cold open and stakeholder pressure.
- 3 15-30: demo target and success criteria.
- 4 30-50: concept burst with worked example.
- 5 50-80: guided practice and prompt clinic.
- 6 80-115: Sprint A builds the first version.
- 7 115-135: failure cards and defect exposure.
- 8 135-165: Sprint B repairs and evidence pack.
- 9 165-180: peer demo, defence note, exit ticket.

Learning outcomes for today

Build

Create the first version of: Baseline comparison, leakage audit, threshold rationale, and one-page decision memo.

Break

Expose at least one agent, data, prompt, model, or workflow failure.

Defend

Explain what can be trusted, what was rejected, and what still needs work.

Team roles

Driver

Operates the repo, files, agent, notebook, tool, or spreadsheet.

Reviewer

Reads outputs, checks assumptions, challenges claims, and watches for defects.

Reporter

Maintains the evidence pack, AI-use log, and defence note.

Retrieval link

Ask the room

- What did we build, break, and defend last week?
- Which habit matters again today?
- Which previous artefact becomes input for this mission?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Cold open discussion

Ask the room

- The business can afford 8,000 retention offers, not 80,000. A model is only useful if it chooses better than a simple rule.
- What would go wrong if the team trusted the agent too early?
- What evidence would make the stakeholder comfortable?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Stakeholder lens

1 Name the stakeholder and the decision they must make.

2 Name the artefact they expect to receive.

3 Name the risk if the artefact is wrong.

4 Name the evidence that would reduce that risk.

5 Name what should be out of scope today.

What makes this an AI-agent course?

Delegation

Students delegate bounded work to the agent instead of doing everything manually.

Supervision

Students inspect, test, reject, and repair agent output.

Accountability

Students defend the final artefact as human analysts, not as agent operators.

Understand the problem before prompting

Students should know the files, definitions, stakeholder, and acceptance criteria before the agent starts.

Starter file walkthrough

- Starter_Kit/sample_data/week3_retention_training.csv
- Sample_Data_and_Scenarios.md cost assumptions
- Starter_Kit/templates/defence_note_template.md
- Portfolio_and_Assessment_Model.md rubric

Confirm before prompting

Ask the room

- What decision will the model change?
- Which features are unavailable at decision time?
- What capacity or budget limit sets the threshold?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Good artefact, weak artefact

Good

Runs or can be inspected, names assumptions, and includes checks.

Weak

Looks polished but hides definitions, dropped records, leakage, or unsupported claims.

Repairable

Imperfect first draft with visible evidence and a clear next fix.

Control the agent before it controls the task

Students practise turning vague requests into supervised work orders.

Week 3

Weak prompt pattern

Too broad

The agent chooses the task boundary and success criteria.

Too trusting

The agent chooses definitions, joins, thresholds, or causal claims.

Too vague

The output cannot be checked because no acceptance criteria were given.

Strong work-order pattern

- 1 Business goal: what decision or workflow does this support?
- 2 Inputs: files, tables, definitions, sources, or prior artefacts.
- 3 Task: the bounded work the agent should perform.
- 4 Acceptance criteria: what must be true before humans accept it.
- 5 Constraints: what the agent must not invent, drop, infer, or overclaim.
- 6 Output format: code, table, memo, test, checklist, or tool.

Activity 1 - prompt rewrite

10 min

Decision frame: action, eligibility, capacity, cost, and value.

Output before moving on

- One revised work order.
- One acceptance criterion.
- One constraint that prevents a known failure.

Prompt rewrite debrief

Debrief prompts

- Which word in the prompt reduced ambiguity?
- Which acceptance criterion can actually fail?
- Which constraint protects the stakeholder?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Concept Depth

Short teaching, immediate use

Use these concept slides to equip the build, not to turn the seminar into a lecture.

Week 3

Concept 1: Baselines before models.

Meaning

How this idea changes today's artefact:
Baseline comparison, leakage audit, threshold rationale, and one-page decision memo.

Failure prevented

Connect it to one failure card: Post-outcome column leaks the answer.

Evidence

Students must show:
Baseline table.

Concept 1 check

Ask the room

- What would the agent do if this concept were ignored?
- What would a strong student artefact include?
- What is the smallest evidence item that proves the concept was used?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Concept 2: Leakage audit before celebration.

Meaning

How this idea changes today's artefact:
Baseline comparison, leakage audit, threshold rationale, and one-page decision memo.

Failure prevented

Connect it to one failure card: Accuracy looks high because churn is rare.

Evidence

Students must show:
Leakage-audit output.

Concept 2 check

Ask the room

- What would the agent do if this concept were ignored?
- What would a strong student artefact include?
- What is the smallest evidence item that proves the concept was used?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Concept 3: Threshold as a business decision.

Meaning

How this idea changes today's artefact:
Baseline comparison, leakage audit, threshold rationale, and one-page decision memo.

Failure prevented

Connect it to one failure card: Agent picks 0.5 threshold without cost logic.

Evidence

Students must show:
Expected-value calculation.

Concept 3 check

Ask the room

- What would the agent do if this concept were ignored?
- What would a strong student artefact include?
- What is the smallest evidence item that proves the concept was used?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Worked Example

Model the evidence standard

Students should imitate the structure of evidence, not copy the example.

Week 3

Worked example highlights

- Good leakage audit: excludes target and post-offer response.
- Good baseline table: compares model with simple business rules.
- Good decision memo: ties threshold to cost, capacity, and limits.

Worked example critique

Ask the room

- What claim is being made?
- Where is the evidence?
- What limit is named?
- What would you ask before accepting it?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Make students do the thinking before the build

The goal is not coverage. The goal is a better first sprint.

Week 3

Exercise 1

12 min

Decision frame: action, eligibility, capacity, cost, and value.

Output before moving on

- A visible table, prompt, test, note, or decision.
- One assumption written down.
- One peer or instructor question answered.

Exercise 1 debrief

Debrief prompts

- What was easy for the agent?
- What decision had to stay human?
- What evidence changed the team's mind?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Exercise 2

12 min

Leakage hunt: label every column by decision-time availability.

Output before moving on

- A visible table, prompt, test, note, or decision.
- One assumption written down.
- One peer or instructor question answered.

Exercise 2 debrief

Debrief prompts

- What was easy for the agent?
- What decision had to stay human?
- What evidence changed the team's mind?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Exercise 3

18 min

Threshold economics: choose a top-N or cutoff policy.

Output before moving on

- A visible table, prompt, test, note, or decision.
- One assumption written down.
- One peer or instructor question answered.

Exercise 3 debrief

Debrief prompts

- What was easy for the agent?
- What decision had to stay human?
- What evidence changed the team's mind?

Instructor listens for

- A specific artefact, not a general intention.
- A named failure mode.
- Evidence that can be inspected.
- A limit or no-use condition.

Sprint A

Build the first version

Time-box the build so students have something real to attack.

Week 3

Sprint A operating rules

- 1 Start from the work order, not a blank chat.
- 2 Keep the agent output visible and reviewable.
- 3 Run the artefact before improving it.
- 4 Write down the first defect or uncertainty.
- 5 Commit or save the first version before repair.

Sprint A task list

- Ask the agent to build three baselines.
- Train one candidate model.
- Compute expected value at multiple thresholds.

Sprint A checkpoint

Ask the room

- What runs or can be inspected right now?
- What did the agent assume?
- What evidence is missing?
- What is the first defect to investigate?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Instructor roaming script

- 1 Ask: show me the artefact.
- 2 Ask: show me what the agent changed.
- 3 Ask: show me one check or known value.
- 4 Ask: what would make this unsafe?
- 5 Do not solve unless the team is blocked by setup rather than thinking.

Break Moment

Failure is the teaching device

The class now attacks the first draft so students learn what not to trust.

Week 3

Failure card 1

Defect

Post-outcome column leaks the answer.

How to expose it

Create a check, adversarial example, missing-value case, source comparison, or peer question.

What to record

Save the failing output before repairing it. This becomes portfolio evidence.

Failure card 2

Defect

Accuracy looks high
because churn is rare.

How to expose it

Create a check,
adversarial example,
missing-value case,
source comparison, or
peer question.

What to record

Save the failing output
before repairing it. This
becomes portfolio
evidence.

Failure card 3

Defect

Agent picks 0.5 threshold without cost logic.

How to expose it

Create a check, adversarial example, missing-value case, source comparison, or peer question.

What to record

Save the failing output before repairing it. This becomes portfolio evidence.

Defect wall

Ask the room

- Post one failure your team found.
- Mark whether it was agent, data, prompt, model, tool, or claim failure.
- Write the evidence that exposed it.

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Sprint B

Repair, harden, and produce evidence

Students now move from one-off fixing to recurrence prevention.

Week 3

Repair moves

- Add forbidden-feature checks.
- Compare against business-rule and random baselines.
- Document the threshold and sensitivity range.

Evidence design checklist

- 1 Name the claim the artefact makes.
- 2 Name the failure that could make the claim false.
- 3 Show the check, test, trace, source, or review that exposes the failure.
- 4 Show what changed after repair.
- 5 State the remaining limit.

Sprint B checkpoint

Ask the room

- Can you reproduce the failure?
- Can you show the fix?
- Would the failure be caught earlier next time?
- What will you put in the evidence pack?

Capture on board

- Definitions students rely on.
- Assumptions that need evidence.
- Risks that should become tests.
- Questions the agent should not decide alone.

Make evidence social

A peer should be able to inspect the artefact and challenge the defence.

Week 3

Peer demo protocol

- 1 One minute: run or show the artefact.
- 2 One minute: show the failure found.
- 3 One minute: show the repair evidence.
- 4 One minute: peer asks what still should not be trusted.
- 5 One minute: team writes one improvement before submission.

Peer review questions

Artefact

What did you build, and can someone else inspect it?

Failure

What did the agent, data, model, tool, or answer get wrong?

Defence

Which claim is supported, and what remains out of scope?

Turn activity into assessable evidence

The defence note is where students show judgement, not just effort.

Week 3

Defence note writing

10 min

The decision policy I can defend is...

Output before moving on

- One claim the team can defend.
- One evidence item linked to that claim.
- One remaining risk or no-use condition.

Evidence pack requirements

- Baseline table.
- Leakage-audit output.
- Expected-value calculation.
- Decision memo with who receives an offer and why.

Rubric lens

Useful artefact

It runs, can be inspected, or clearly solves the mission within scope.

Agent supervision

The team shows what was accepted, rejected, changed, and verified.

Defensible judgement

Claims are specific, evidence-linked, and include limits.

Close

Make the habit portable

Students should leave with one artefact, one failure, and one defensible claim.

Week 3

Common traps to avoid

- Optimising AUC while ignoring capacity.
- Letting the agent choose the target window.
- Reporting a model score without a decision policy.

Stretch options for fast teams

- Add fairness or service-quality slices.
- Compare model selection with a simple tenure-based rule.

After-class handoff

- 1 Commit or save the artefact.
- 2 Save the failed or rejected output.
- 3 Complete the AI-use log.
- 4 Submit the defence note.
- 5 Bring one unresolved question to the next retrieval link.

Agent work order

Give the agent a job, not a wish.

Use this structure

- Goal: rank customers for a limited retention offer.
- Inputs: checked Week 2 data and cost assumptions.
- Acceptance: at least three baselines, no post-outcome features, threshold justified by cost.
- Log every model run with data version and prompt reference.

Studio sprint A - build the first version

Tasks

- Ask the agent to build three baselines.
- Train one candidate model.
- Compute expected value at multiple thresholds.

Instructor watchlist

- Is the prompt specific enough?
- Did the team run the artefact?
- Did the team record what the agent changed?

Failure cards

Release these after Sprint A. Students must expose the defect before fixing it.

Card 1

**Post-outcome
column leaks the
answer.**

Card 2

**Accuracy looks high
because churn is
rare.**

Card 3

**Agent picks 0.5
threshold without
cost logic.**

Studio sprint B - repair and harden

Repair moves

- Add forbidden-feature checks.
- Compare against business-rule and random baselines.
- Document the threshold and sensitivity range.

Definition of done

- The failure is reproducible.
- The fix is committed.
- A future failure would be caught earlier.

Evidence pack

What students should submit

- Baseline table.
- Leakage-audit output.
- Expected-value calculation.
- Decision memo with who receives an offer and why.

Peer demo

Demo in three minutes

- Run the artefact.
- Show the failure you found.
- Show the evidence that supports the final claim.

Peer questions

- What did the agent get wrong?
- What would fail next?
- What should not be used yet?

Common traps

Watch for these

- Optimising AUC while ignoring capacity.
- Letting the agent choose the target window.
- Reporting a model score without a decision policy.

If your team finishes early

Stretch options

- Add fairness or service-quality slices.
- Compare model selection with a simple tenure-based rule.

Exit ticket

The decision policy I can defend is...

Write one sentence. It goes into your portfolio evidence.

Leave with this

**A model becomes useful only when it
changes a constrained decision.**

BUILD / BREAK / DEFEND